INCH-POUND

MIL-B-29176B(YD)
27 February 1990
SUPERSEDING
MIL-B-29176A(YD)
13 January 1984

MILITARY SPECIFICATION

BUILDING, PREFABRICATED, STEEL ARCH, DECONTAMINATION CENTER

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one size of prefabricated, ready-cut, steel, trussless arch building complete with entrance tunnel, emergency exit structure, blast door, flap gate, interior steel doors and steel partitions, steel ladder, anchor bolts, endwalls, 1-inch diameter steel rods, clevises, turn buckles and eyebolts, base channels, connector angles and hardware, and all exhaust and air intake pipes and fittings. The prefabricated building shall be easily assembled.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5410

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SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-T-97 - Tape, Packaging/Industrial, Filament Reinforced.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-C-3774 - Crate, Wood, Open, 12,000 and 16,000 Pound (lb) Capacity.
MIL-C-52950 - Crates, Wood, Open and Covered.

STANDARDS

MILITARY

MIL-STD-2073 - DoD Material, Procedures of Development and Application of Packaging Requirements.

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Military Specifications and Standards, Bldg., 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Government drawing. The following Government drawing form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWING

NAVAL FACILITIES ENGINEERING COMMAND

1355604 - Decontamination Center, Architectural.

(Copies of specifications, standards, drawings and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents which are current on the date of the solicitation (see 6.2).

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

Specification for the Design, Fabrication, and Erection of Structural Steel Buildings.

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(Application for copies should be addressed to the American Institute of Steel Construction, 101 Park Avenue, New York, NY 10017.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

ANSI - A14.3 - Safety Requirements for Fixed Ladders.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

ASTM

ASTM A 153	- Zinc Coating (Hot-Dip) on Iron and Steel Hardware, Specification for.
ASTM A 525	- Steel Sheet, Zinc-Coated (Galvanized) by the Hot-Dip Process, General Requirements, Specification for.
ASTM B 633	- Electrodeposited Coatings of Zinc on Iron and Steel, Specification for.
ASTM A 760/A 760M	- Pipe, Corrugated Steel, Zinc-Coated (Galvanized), Specification for.
ASTM D 3951	- Commercial Packaging.
ASTM D 3953	- Flat Steel Strapping.

(Application for copies should be addressed to the ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. Each prefabricated 25-foot wide by 68-foot long circular steel arch building shall be furnished in accordance with Drawing 1355604 and as specified herein.

3.2 First article. When specified (see 6.2), the contractor shall furnish a steel arch prefabricated building for first article inspection and approval (see 4.2.1 and 6.4).

3.3 Materials. Materials shall be as specified herein and in applicable specifications and standards, and other referenced documents. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Materials shall be free of defects which adversely affect performance or serviceability of the finished product.

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3.3.1 Steel.

3.3.1.1 Structural steel. Steel for structural members, blast door, flap gate, steel doors, and steel partitions shall conform to section 1.4.1 of AISC specification for the Design, Fabrication and Erection of Structural Steel for Buildings.

3.3.1.2 Sheet steel. Sheet steel shall have a minimum yield stress of 33,000 pounds per square inch and shall conform to ASTM A 525. Rib configuration of the sheets shall be in accordance with the manufacturers standard practice.

3.3.1.3 Zinc coating. The zinc coating for sheet steel shall conform to ASTM A 525, coating designation G90. Zinc coating for fasteners shall conform to ASTM A 153, class C or D as applicable, or ASTM B 633, type CG.

3.3.2 Corrugated pipe. Corrugated pipe shall conform to ASTM A 760/A 760M.

3.3.3 Ladder. The steel ladder shall conform to ANSI A14.3.

3.4 Construction. The building shall be fabricated to assure ease of assembly, fit, interchangeability of like components, watertightness, and proper operation of doors, exit hatch, and exhaust and intake air pipes and fittings. A sealant system shall be provided with each building. The contractor is responsible for providing in accordance with industry standards for similar buildings to insure proper fit, assembly, and operation of the building components.

3.5 Markings.

3.5.1 Erection marks. The erection mark of each component, structural member, or piece shall be stenciled on the item or stamped on an attached metal tag in the case of rods or other narrow items. Markings shall be Gothic type capitals and Arabic numerals not less than 1-inch high, except markings on attached metal tags shall be not less than 3/8-inch high. Stenciling, when used, shall be applied with a waterproof paint.

3.6 Workmanship.

3.6.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.6.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.6.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not

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countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.6.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.7 Treatment and painting. Unless otherwise specified (see 6.2), the building shall be treated and painted in accordance with the manufacturer's standard practice. All surfaces of the building other than corrosion-resisting steel shall be protected against corrosion and present a neat appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Material inspection. The contractor is responsible for ensuring that supplies and materials are inspected for compliance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

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4.2.1 First article inspection. The first article inspection shall be performed on one complete building when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.3. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3 and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3. Examination. Each building selected shall be examined for compliance with the requirements in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirement or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Packaging inspection. The preservation, packing, and marking of the item shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1. Preservation. The preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Methods of preservation. Cleaning processes, drying procedures, preservations, and methods of preservation are listed in MIL-P-116 and shall conform to the requirements of MIL-P-116 and any applicable specifications.

5.1.1.2 Cleaning and drying. Prior to the application of preservative compounds or paint, surfaces shall be cleaned by process C-1 and dried by any applicable procedure of MIL-P-116.

5.1.1.3 Unpainted surfaces. Unpainted and uncoated ferrous metal surfaces of component parts of the building, shall be coated with type P-1 preservative. Joints of hinges shall be coated with type P-10 preservatives.

5.1.1.4 Paint and calking. Paint and calking compound, for one complete building, shall be packaged in fiberboard boxes conforming to PPP-B-636, class weather resistant.

5.1.1.5 Fasteners. Fasteners for one complete building shall be packaged method III.

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5.1.1.6 Ventilators, louvers, smokestacks, and equipment door hardware. The items needed for one complete building shall be packaged method III.

5.1.1.7 Flashing, ducts, beam and girder connectors, struts, braces, rods, roof tie assemblies, and ridge end covers. These items of like description for one complete building shall be bundled together to form a nonshifting bundle which will provide convenient handling. The bundles shall be secured with tape conforming to PPP-T-97, type IV, or with soft annealed wire. The threaded surface of the rods shall be protected to prevent damage to the threads.

5.1.1.8 Technical publications. The technical publications required for each building shall be preserved method IC-3.

5.1.1.9 Consolidated packaging. Technical publications and other small components for one complete building requiring the protection of a container shall be consolidated and packaged in boxes conforming to PPP-B-636, class weather resistant. The contents shall be cushioned, blocked, and braced to prevent movement.

5.1.2 Commercial. The complete building shall be preserved in accordance with ASTM D 3951.

5.2 Packing. The packing shall be level A or commercial as specified (see 6.2).

5.2.1 Level A. Only one complete unassembled building shall be packed as a set consisting of bundled boxes and crates. Components shall be arranged within each bundle, box, or crate to provide ease of erection of the building when removed. Sequence of packing shall conform to sequence of construction as nearly as possible, so all items needed for a particular assembly and operation are packed together, obviating the necessity to open, search, and expose the contents of particular bundles or containers to the weather merely to locate a few needed items.

5.2.1.1 Doors, louvers, smokestacks, bundled components, and consolidated packages. These items shall be packed in accordance with MIL-STD-2073. Woodboxes shall be selected from the level A containers in table VII of MIL-STD-2073.

5.2.1.2 Structural members. Structural members, such as steel girders, purlins, and similar components shall be nested, and secured with bolts or steel straps, or a combination of both, to form compact nonshifting bundles. Suitable wood blocking shall be used, as required, under strapping to eliminate large voids or irregular shaped bundles. Strapping shall be 0.023-inch by 3/4-inch flat steel conforming to ASTM D 3953. Strapping shall be stapled to any wood blocking provided. Strapping shall be spaced not to exceed approximately 36 inches on center, with end straps placed not more than 18 inches from each end. Metal edge protectors shall be used to avoid excess strapping pressure on edges of structural members in the bundles. Bundled structural members shall have wood secured to the bottom of the load to provide for handling by forklift and for stacking. The height of the bottom shall be not less than 2-1/1 inches. The bundles shall not exceed approximately

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5,000 lbs. The individual bundles shall be packed to provide minimum movement and shall be blocked, braced, or otherwise anchored in accordance with the appendix to the crate specification.

5.2.1.3 Sheeting. Roof and wall sheeting shall be packed in crates conforming to MIL-C-52950, type II or III, or MIL-C-3774, type I or II depending upon the size and weight of each load. Contents shall be waterproofed (with full shrouds), cushioned, anchored, and closed in accordance with the appendix to the applicable crate specification.

5.2.2 Commercial. The complete building shall be packed in accordance with ASTM D 3951.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1. Intended use. The buildings are intended to be used as decontamination centers where up to 40 people can live in the event of nuclear attack.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required for inspection and approval (see 3.2, 4.2.1, and 6.4).
d. When treatment and painting are required (see 3.7).
e. Level of preservation and level of packing required (see 5.1 and 5.2).

6.3 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD Federal Acquisition Regulation (FAR) Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data should be delivered by the contractor in accordance with the contract or purchase order requirements.

6.4 First article. When a first article inspection is required, (see 3.2, 4.2.1, and 6.2) it shall be tested and approved under the appropriate provision of paragraph 7-104.55 of the Armed Services Procurement Regulation (ASPR). The first article shall consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

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6.5 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. The unit of product shall be one prefabricated building. All buildings offered for delivery at one time shall be considered a lot for the purpose of inspection. The inspection level shall be level S-2 and the Acceptable Quality Level (AQL) shall be 4.0 percent defective. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be preformed for all defects. Rejected lots shall be separate from new lots, and shall be clearly identified as reinspected.

6.6 Subject term (key word) listing.

Prefabricated building.
Steel arch building.
Trussless arch building.

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Navy - YD

Preparing Activity:
Navy - YD

(Project 5410-N333)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-B-29176B	900227

3. DOCUMENT TITLE
BUILDING, PREFABRICATED, STEEL ARCH, DECONTAMINATION CENTER

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
COMMANDING OFFICER	805-982-5572	551-5572
c. ADDRESS (Include Zip Code) NAVAL CONSTRUCTION BATTALION CENTER CODE 156 PORT HUENEME, CA 93043-5100	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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